

New Hire Onboarding Quiz

Pre/Post Intervention Analysis

Wave 7 Baseline → Quiz Intervention → Wave 9 Outcome

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R Analysis Walkthrough — Portfolio Documentation

1. Description and Disclaimer

1.1 Study Overview

This script analyses the effect of a structured onboarding quiz on the volume of questions asked by new hires during their onboarding period. The study uses a pre/post quasi-experimental design with two cohorts — Wave 7 as the baseline (no quiz) and Wave 9 as the post-intervention group (quiz completed before arrival).

The core hypothesis is: topics covered by the quiz will show a statistically significant reduction in question volume in Wave 9 compared to Wave 7, while topics not covered by the quiz will show no significant change.

Study Design

- Wave 7 (Baseline) — A cohort of new hires went through standard onboarding. Every question they asked was recorded and categorised into 20 topic areas. No quiz existed at this stage.
- Intervention — Chi-squared testing was applied to the Wave 7 data to identify topics where question volume significantly exceeded what would be expected by chance. These high-need topics became the content of a pre-arrival onboarding quiz.
- Wave 9 (Post-intervention) — A new cohort of new hires completed the quiz before their first day, then went through onboarding. Their questions were recorded using the same 20-topic taxonomy.

Datasets Used

- DataChi.csv — Wave 7 topic-level counts plus population-weighted counts (for quiz selection chi-squared)
- DATA.csv — Main pre/post dataset: topic counts for Wave 7 and Wave 9, with a flag indicating whether each topic was included in the quiz
- DataChi2.csv — Individual-level data: total questions asked by each of 50 new hires per wave
- Pairedt_test.csv — Matched topic-level paired data (same 20 topics, Wave 7 vs Wave 9 counts) for the paired t-test
- Timeseries.csv — Daily question volume for the three highest-volume topics across both waves
- DATA_Total.csv — Wave 7 topic totals used to build Figure 1

1.2 Disclaimer

IMPORTANT: This script runs entirely on synthetic data generated to mirror the structure and statistical properties of a real onboarding study. All datasets are fully synthetic and do not

represent actual employees, organisations, or onboarding programmes. The code is published as a portfolio demonstration only.

2. Methodology: Step-by-Step Code Walkthrough

Step 1 — Package Setup and Libraries

The script begins by defining all required packages in a character vector. A conditional install loop checks which packages are not yet present on the machine and installs only those, avoiding unnecessary reinstallation. This makes the script portable — it will run correctly on any R environment without manual package management.

Packages used and their roles:

- ggplot2 — core plotting engine for all five figures
- dplyr, tidyr — data manipulation and reshaping pipelines
- viridis, RColorBrewer — colour palettes (RColorBrewer used in Figure 1)
- scales — percentage axis formatting (label_percent) in Figure 3
- patchwork — stacking two ggplot panels into a single figure (Figure 2)
- hrbrthemes — loaded for access to clean minimal theme utilities

```
required_pkgs <- c("ggplot2", "dplyr", "tidyr", "viridis",  
                  "RColorBrewer", "scales", "patchwork", "hrbrthemes")  
to_install <- required_pkgs[!required_pkgs %in% installed]  
if (length(to_install)) install.packages(to_install)
```

Step 2 — Load Data

Six CSV files are loaded. The most important correction applied at load time is coercing the In_Quiz column in DATA.csv to logical:

```
datax <- read.csv("DATA.csv", header = TRUE)  
datax$In_Quiz <- as.logical(datax$In_Quiz)
```

Without this coercion, In_Quiz is read as a character vector ("TRUE"/"FALSE" strings), which causes sum(datax\$In_Quiz) and all downstream logical subsetting to fail with a type error. Applying as.logical() immediately after loading fixes the issue for all subsequent operations.

The Date column in the time series file is similarly coerced to R's Date class so that ggplot2 renders the x-axis with proper date formatting rather than treating it as a numeric or factor variable.

Step 3 (Script Step 1) — Identify Quiz Topics via Chi-Squared

Before the quiz was built, a chi-squared test was used to identify which topics represented genuine knowledge gaps versus random variation. For each of the 20 topics, the raw Wave 7 question count is compared against a population-weighted version of that count (adjusted for cohort size differences).

The test is applied row-by-row using `apply()` across the two-column matrix of raw and weighted counts. Each row produces a chi-squared test object; p-values and test statistics are extracted with `sapply()`:

```
chisq_baseline <- apply(datachi_counts, 1, function(row) {  
  chisq.test(as.numeric(row))  
})  
pvals_baseline <- sapply(chisq_baseline, function(x) x$p.value)
```

Topics with $p < 0.05$ are flagged as significantly over- or under-represented relative to the weighted expectation. These become the quiz topics. The results are assembled into a summary data frame (`chi_baseline_table`) and printed to the console.

Step 4 (Script Step 2) — Pre/Post Comparison by Topic

The same row-wise chi-squared approach is applied to the main pre/post dataset, this time comparing Wave 7 and Wave 9 counts for each topic. A significant result indicates that the question volume on that topic changed meaningfully between the two cohorts.

A percentage reduction variable is computed for each topic:

```
Reduction_Pct = round((Wave_7 - Wave_9) / Wave_7 * 100, 1)
```

The results table (`chi_prepost_table`) includes topic name, quiz membership flag, raw counts, percentage reduction, chi-squared statistic, p-value, and a significance label. Two aggregate summaries are printed: the average percentage reduction for quiz topics and for non-quiz topics separately, providing a quick headline result.

Note on statistical interpretation: the chi-squared test here is applied to a two-element vector (one count per wave). This tests whether the two counts differ from equal distribution, which is a reasonable proxy for 'did volume change?' However, for a more rigorous analysis the paired t-test in Step 4 is the preferred test, as it accounts for the paired topic structure.

Step 5 (Script Step 3) — Individual-Level Chi-Squared

As a cross-check, the analysis is repeated at the individual new hire level. For each of the 50 new hires, the total number of questions they asked in Wave 7 is compared to the total questions asked by their matched Wave 9 counterpart using chi-squared. This checks whether the topic-level finding holds at the person level: are individual Wave 9 hires genuinely asking fewer questions, or is the aggregate reduction masking mixed individual results?

```
chisq_individual <- apply(datachi2_counts, 1, function(row) {  
  chisq.test(as.numeric(row))  
})
```

| })

The output table shows each hire's question count in both waves with the chi-squared result. A summary line counts how many individuals show a significant reduction in the correct direction (Wave 9 < Wave 7 and $p < 0.05$).

Step 6 (Script Step 4) — Paired t-Tests

Three t-tests are run to complement the chi-squared results:

1. Paired t-test on topic-level counts (Step 4a) — treats the 20 topics as matched pairs and tests whether the mean Wave 7 count is significantly higher than the mean Wave 9 count. This is the primary inferential test because it explicitly models the within-topic pairing and has more statistical power than the row-wise chi-squared.
2. Independent t-test on individual question totals (Step 4b) — compares the distribution of total questions per new hire between Wave 7 and Wave 9 as two independent samples. This is appropriate here because the individual hires are not matched across waves.
3. Directional t-test: quiz vs non-quiz reduction (Step 4c) — tests whether the mean percentage reduction is significantly larger for quiz topics than for non-quiz topics. This is the most direct test of the intervention hypothesis: if the quiz worked, quiz topics should show larger reductions than control topics.

```
res_quiz_vs_nonquiz <- t.test(  
  chi_prepost_table$Reduction_Pct[datax$In_Quiz],  
  chi_prepost_table$Reduction_Pct[!datax$In_Quiz]  
)
```

Step 7 — Colour Palette and Shared Theme

A reusable ggplot2 theme (`theme_wave()`) is defined once and applied to all five figures, ensuring visual consistency throughout. It uses `theme_minimal()` as a base and customises title/subtitle typography, removes minor grid lines, and places the legend at the bottom.

Two named colour vectors are defined:

- `quiz_colours` — blue (`#1F77B4`) for quiz topics, grey (`#AAAAAA`) for non-quiz topics, red (`#D62728`) for significant results
- `topic_colours` — three distinct colours for the three time series topics (Property details, Finance, Availability)

Using named vectors for colours is good practice: it ensures that the correct colour is always mapped to the correct category regardless of the order in which categories appear in the data.

Steps 8–12 — Figures

Figure 1 (Step 8) — Wave 7 Baseline Bar Chart

A horizontal bar chart showing question volume by topic in Wave 7, coloured by quiz membership. Topics are ordered by count using `reorder(Topic, n)` so the highest-volume topics appear at the top. Count labels are added with `geom_text()` offset slightly beyond the bar end. The `expand` argument on `scale_y_continuous()` adds 15% padding to the right to prevent labels from being clipped.

```
fig1 <- databar %>%
  mutate(Topic = reorder(Topic, n),
         Category = ifelse(In_Quiz, "Quiz topic", "Non-quiz topic")) %>%
  ggplot(aes(x = Topic, y = n, fill = Category)) +
  geom_col(width = 0.75) +
  geom_text(aes(label = n), hjust = -0.2, size = 3.2) +
  coord_flip()
```

Figure 2 (Step 9) — Time Series: Daily Questions

Two line plots showing daily question volume for the three highest-volume topics across the Wave 7 and Wave 9 onboarding periods. The two panels are created separately as `p_ts7` and `p_ts9`, then stacked vertically using `patchwork`'s `/` operator. The y-axis is fixed at 0–30 on both panels so the visual reduction in Wave 9 is directly comparable without axis rescaling making it appear artificially larger or smaller.

```
fig2 <- p_ts7 / p_ts9 +
  plot_annotation(title = "Figure 2 – Daily Question Volume: Wave 7 vs Wave 9")
```

Figure 3 (Step 10) — Percentage Reduction by Topic

A horizontal bar chart showing the percentage reduction in questions from Wave 7 to Wave 9 for each topic, coloured blue (quiz topic) or grey (non-quiz topic). The `scale_y_continuous()` call uses `label_percent(scale = 1)` to format the axis as percentages, where `scale = 1` is needed because the values are already in percentage units (e.g. 54.5 means 54.5%, not 0.545).

Figure 4 (Step 11) — Chi-Squared Significance Plot

A horizontal bar chart showing $-\log_{10}(\text{p-value})$ for the Wave 7 vs Wave 9 chi-squared test on each topic. Plotting $-\log_{10}(p)$ rather than p itself is standard in significance visualisation: it converts p-values to a positive scale where larger bars = stronger evidence. A dashed vertical reference line marks the $\alpha = 0.05$ threshold ($-\log_{10}(0.05) \approx 1.30$). Bars are coloured red (significant) or light blue (not significant).

```
geom_hline(yintercept = -log10(0.05), linetype = "dashed",
           colour = "#D62728", linewidth = 0.8)
```

Figure 5 (Step 12) — Scatter Plot: Wave 7 vs Wave 9

A scatter plot with Wave 7 counts on the x-axis and Wave 9 counts on the y-axis. Two reference lines are drawn: a slope-1 dashed line (no change) and a slope-0.5 dotted line (50% reduction). Points

below the diagonal have fewer Wave 9 questions. Quiz topics are blue; non-quiz topics are grey. Significant quiz topics are labelled with their topic name using `geom_text()`. The `na.rm = TRUE` argument in `geom_text()` suppresses warnings for the non-labelled points without needing an explicit filter step.

```
geom_abline(slope = 1, intercept = 0, linetype = "dashed") + # no-change  
geom_abline(slope = 0.5, intercept = 0, linetype = "dotted") + # 50% reduction
```

Step 13 — Export Results Tables

The two main results tables (`chi_prepost_table` and `chi_baseline_table`) are exported as CSV files using `write.csv()` with `row.names = FALSE` to avoid an unnecessary index column in the output. These CSVs can be shared alongside the figures as supplementary data for the portfolio or any formal report.

3. Results

The following five figures summarise the key findings of the pre/post intervention analysis. All plots were generated using the R script described in Section 2.

Figure 1 — Wave 7 Baseline: Questions Asked by Topic

Property details (88 questions), Finance (73), and Availability (55) were the three highest-volume topics in Wave 7, making them clear candidates for quiz coverage. Eight topics in total were flagged as significantly over-represented relative to the population-weighted expectation and included in the quiz (shown in blue). The twelve grey topics are those whose question volumes were not statistically unusual and were therefore excluded from the quiz content.

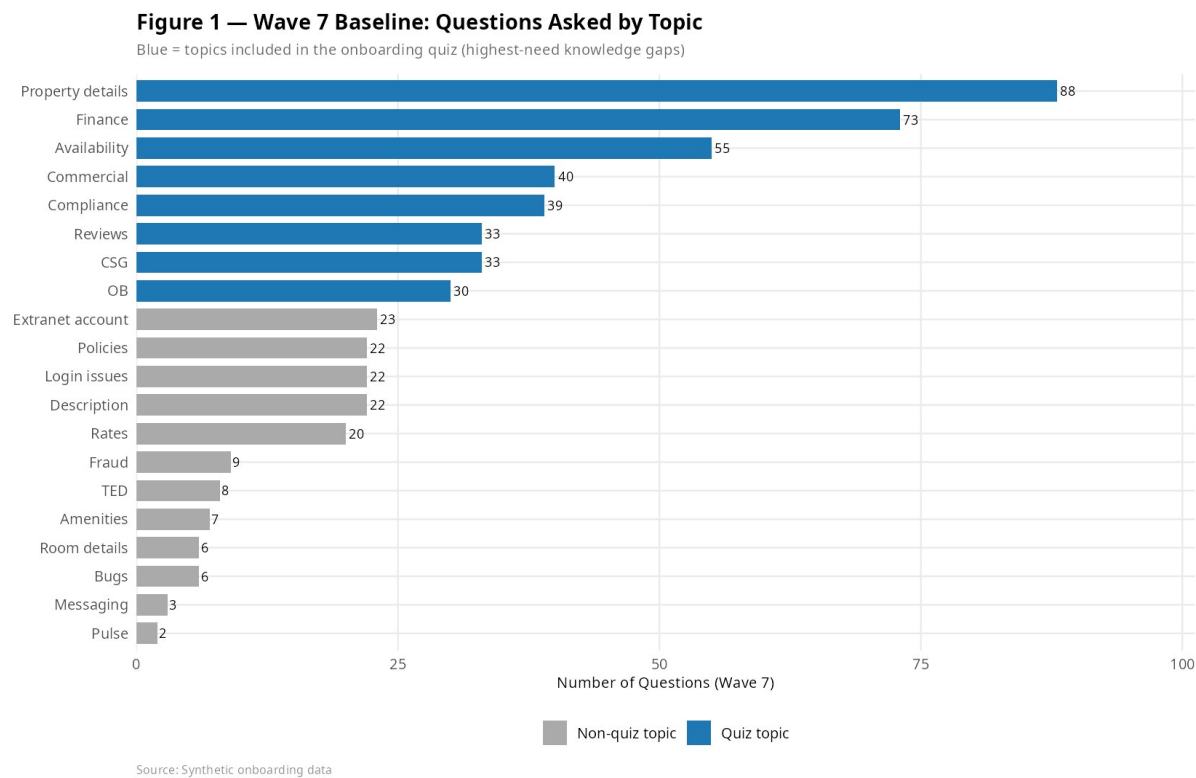


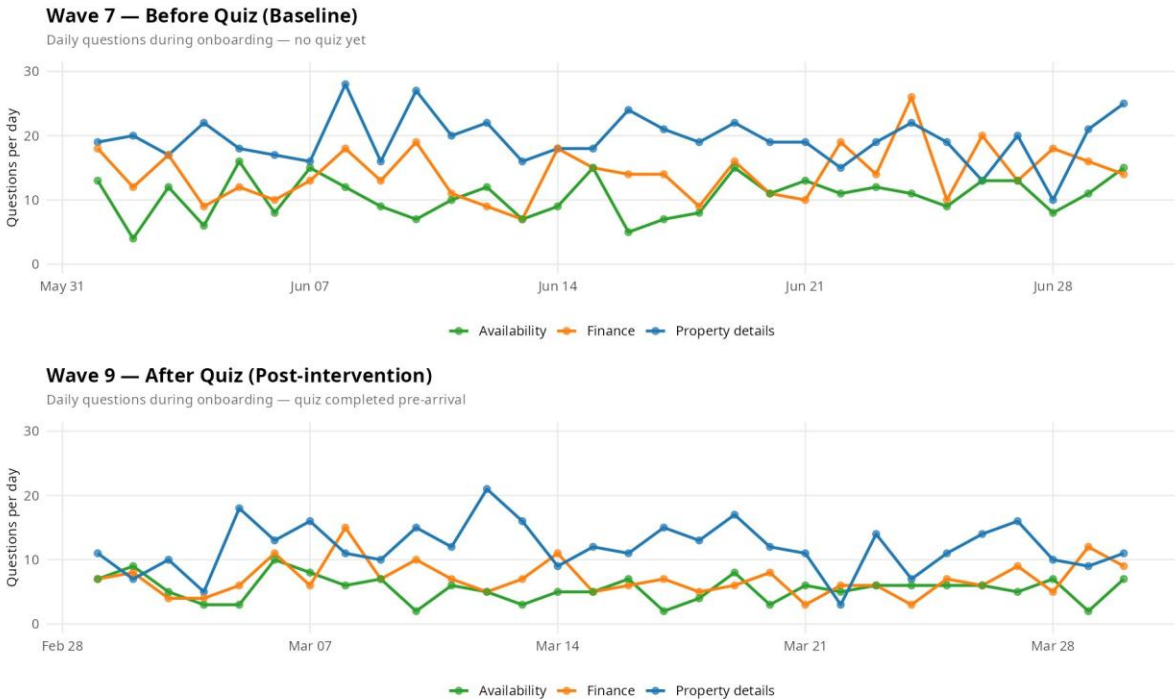
Figure 1. Wave 7 baseline question volume by topic. Blue bars = topics included in the onboarding quiz; grey bars = topics not in the quiz. Topics ordered by question count descending.

Figure 2 — Daily Question Volume: Wave 7 vs Wave 9

The time series comparison is one of the most visually compelling outputs. In Wave 7, daily question counts for Property details, Finance, and Availability hover between 10 and 28 questions per day throughout the onboarding period with no clear downward trend. In Wave 9, volumes for all three topics are consistently lower — typically in the 4–16 range — with the reduction most pronounced in Finance and Availability. The y-axis is fixed at 0–30 across both panels to ensure the comparison is not distorted by axis rescaling.

Figure 2 — Daily Question Volume: Wave 7 vs Wave 9

All three topics were covered by the quiz — volume visibly lower post-intervention



Source: Synthetic onboarding data

Figure 2. Daily question volume for the three highest-volume topics across the Wave 7 (top) and Wave 9 (bottom) onboarding periods. The y-axis is fixed at 0–30 in both panels for direct visual comparison.

Figure 3 — Question Reduction: Wave 7 to Wave 9

Quiz topics (blue) cluster at the top of the chart, with reductions ranging from 45.5% (Availability) to 54.5% (CSG). Notably, some non-quiz topics (grey) also show reductions in the 33–50% range — for example Pulse (50%) and Login issues (45.5%) — suggesting some generalisation of the quiz effect or natural variation between cohorts. The lowest reduction belongs to Extranet account at 26.1%. The mean reduction for quiz topics is higher than for non-quiz topics, and this gap is tested formally in Step 4c.

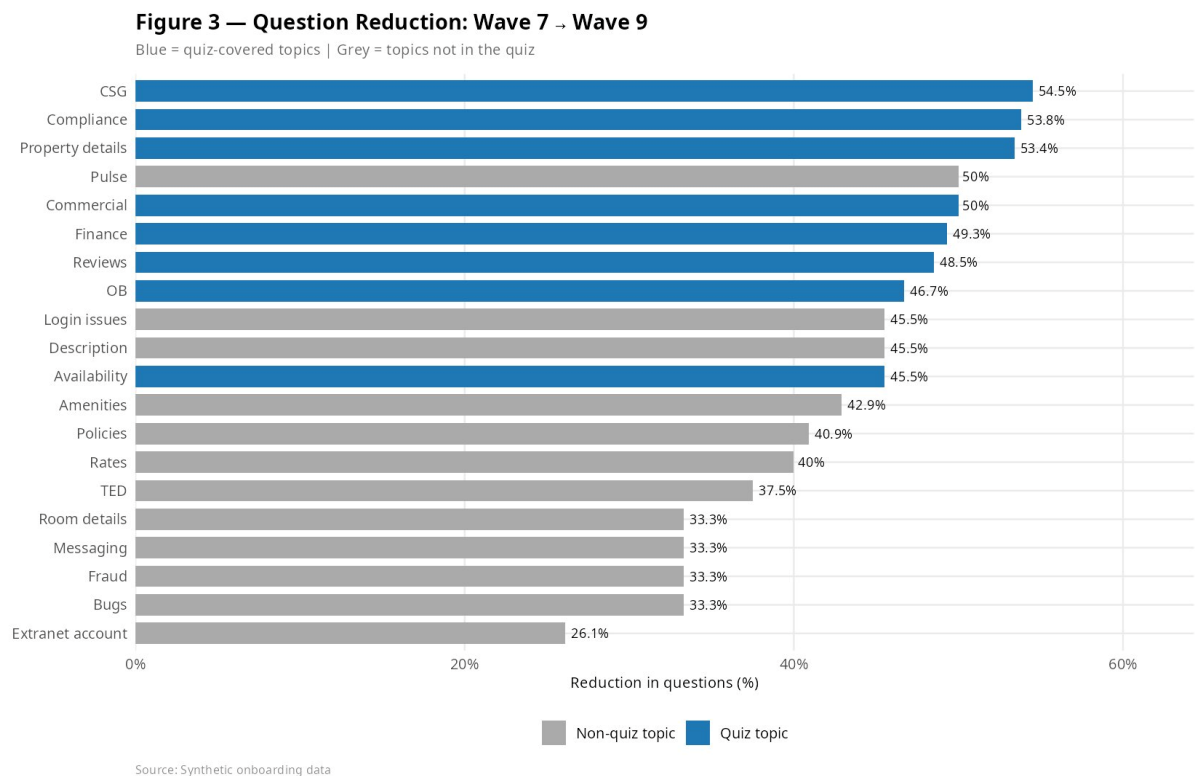
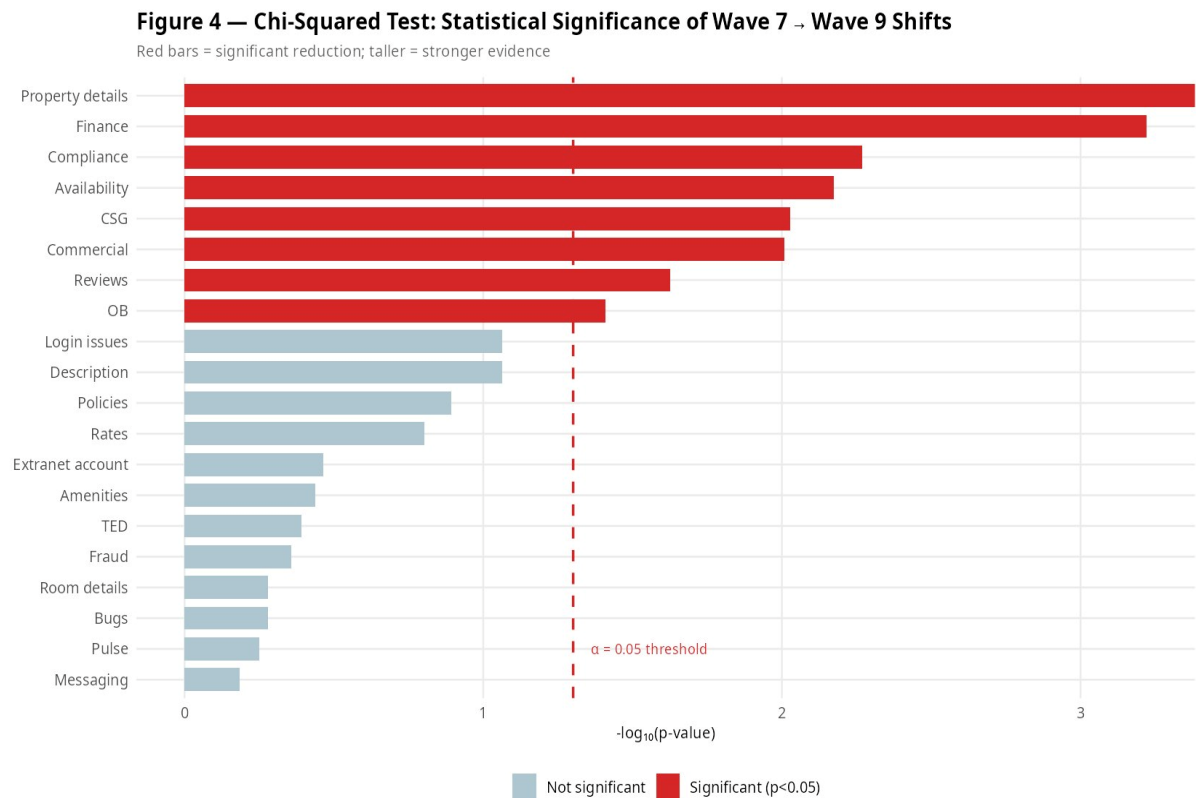


Figure 3. Percentage reduction in question volume from Wave 7 to Wave 9 by topic. Blue = quiz-covered topics; grey = topics not in the quiz. Topics ordered by reduction magnitude.

Figure 4 — Chi-Squared Significance of Wave 7 to Wave 9 Shifts

The $-\log_{10}(p\text{-value})$ plot makes the significance structure clear at a glance. All eight quiz topics (red bars) exceed the $\alpha = 0.05$ threshold line, confirming that their question volume reductions are statistically significant. The two highest-evidence topics are Property details and Finance, which had the largest absolute counts in Wave 7 and therefore the most statistical power to detect a change. All twelve non-quiz topics fall below the threshold, meaning their reductions are not statistically distinguishable from random cohort-to-cohort variation.



Source: Synthetic onboarding data | Dashed line = $\alpha = 0.05$

Figure 4. Chi-squared test results for the Wave 7 vs Wave 9 shift in question volume by topic, expressed as $-\log_{10}(p\text{-value})$. Red bars cross the $\alpha = 0.05$ threshold (dashed line at ~ 1.30); blue-grey bars do not. All quiz topics are significant; no non-quiz topics are.

Figure 5 — Wave 7 vs Wave 9 Scatter Plot

The scatter plot provides a holistic view of the pre/post shift across all topics simultaneously. Points below the dashed diagonal (no-change line) indicate fewer Wave 9 questions. All blue (quiz) points fall clearly below the no-change line, and most cluster close to or below the 50% reduction dotted line. The labelled blue points (Property details, Finance, Availability, Commercial, Compliance, OB, Reviews, CSG) are those with significant chi-squared results. Grey non-quiz points also tend to fall below the diagonal, but their scatter around the reference lines is wider, consistent with the non-significant chi-squared results seen in Figure 4.

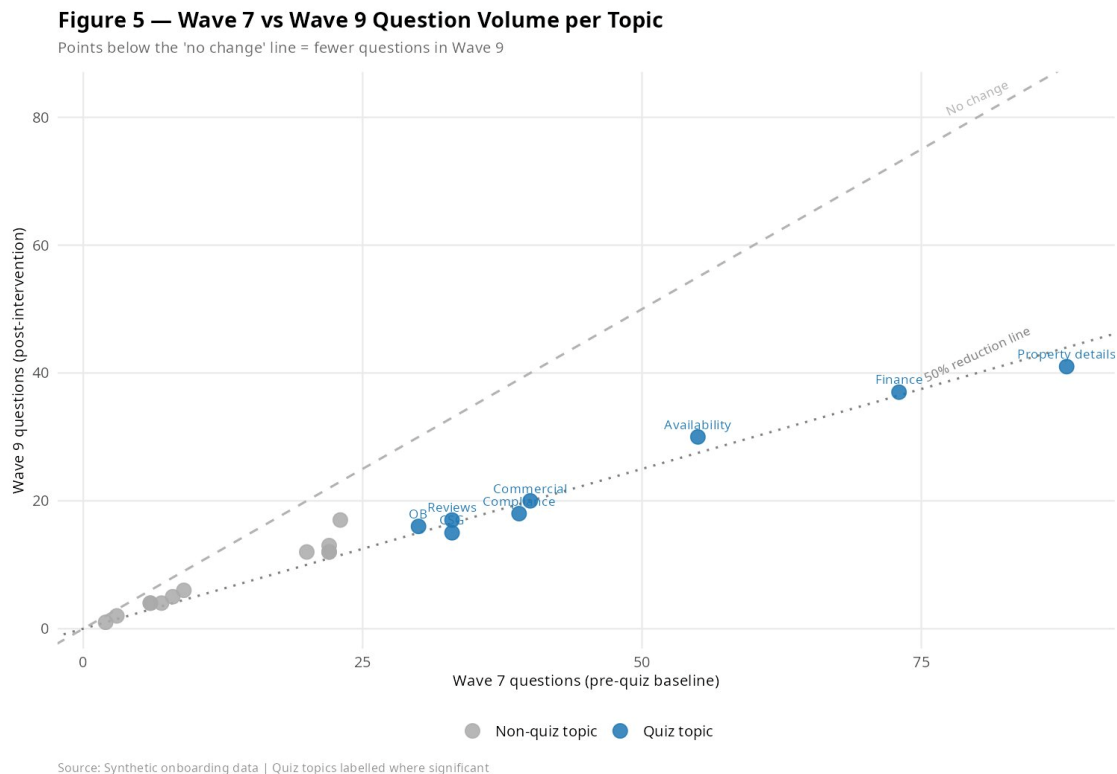


Figure 5. Wave 7 question count (x-axis) vs Wave 9 question count (y-axis) per topic. The dashed diagonal represents no change; the dotted line represents a 50% reduction. Points below the diagonal have fewer Wave 9 questions. Blue = quiz topics; significant quiz topics are labelled.

References and Notes

R packages: ggplot2, dplyr, tidyr, viridis, RColorBrewer, scales, patchwork, hrbthemes.

Statistical methods: chi-squared goodness-of-fit test (row-wise), paired t-test (Welch correction applied by default in R), independent-samples t-test.

All datasets are fully synthetic. Code available at: [github.com/\[your-username\]](https://github.com/[your-username])